

Pre-Board Examination - 2079

Class : XII

Subject: Biology (Code: 2021)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet.

[5 × 1 = 5]

1. Fruit ripening hormone is
 - a. Ethylene
 - b. Auxin
 - c. Gibberellins
 - d. Cytokinins
2. The following plant has hypo-stomatic leaf
 - a. Monocot
 - b. Dicot
 - c. Mango
 - d. Banyan
3. A pollination that occurs between the flowers of different branches of the same plant is called
 - a. Self-pollination
 - b. Cross-pollination
 - c. Geitonogamy
 - d. Cleistogamy
4. The process of finding out the age of a tree by counting the number of annual rings is called
 - a. Phytology
 - b. Dendrology
 - c. Analogy
 - d. Dendrochronology
5. The genes associated to y-chromosomes are called
 - a. Holandric genes
 - b. Sex-linked genes
 - c. Androgens
 - d. X-linked genes

Group 'B' [BOTANY]

Give short answer to the following questions.

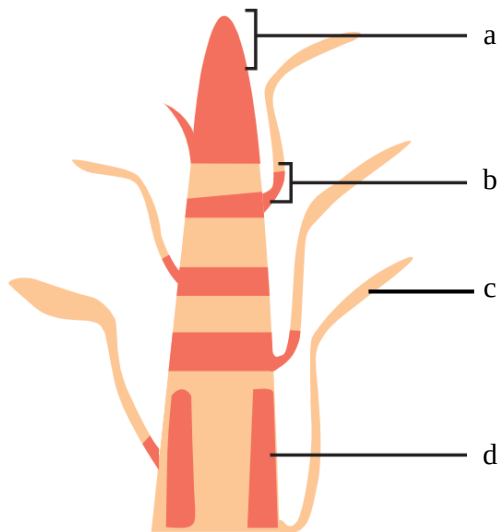
[4 × 4 = 16]

1. What is cross pollination? Mention the advantages of cross pollination. [1+3 = 4]
2. Define genetic engineering. State the dangers associated to this technology? [1+3 = 4]
3. What is meristematic tissue? Mention its characteristics. [1+3 = 4]

OR

Distinguish between photosynthesis and respiration.

4. Name the parts a, b, c and d. [4]



Group 'C' [BOTANY]

Give long answer to the following questions.

[2 × 8 = 16]

5. What do you mean by sex-linked inheritance? Why a father cannot transfer X-linked gene to his sons? Show the inheritance pattern of colour-blindness in human.

[1+1+6 = 8]

OR

Explain the semi-conservative method of DNA replication

6. Name the end product of glycolysis. Explain Krebs's cycle. [1+7 = 8]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet.

[6 × 1 = 6]

1. In mammalian skin adipose tissue is found
 - a. In muscles
 - b. In the dermis
 - c. Below the dermis
 - d. In the epidermis
2. Animal pole in frog's egg in water remains upwards because:
 - a. Sperms enter through it
 - b. It absorbs light
 - c. It is free from yolk
 - d. It makes egg invisible to enemies by absorbing light
3. Bruner's glands are related to
 - a. Digestive system
 - b. Sweating system
 - c. Male reproductive system
 - d. Female reproductive system
4. Urea is produced from ammonia in the body of rabbit or man in
 - a. Liver
 - b. Kidneys
 - c. Urinary bladder
 - d. Blood
5. The hormone ADH is secreted by
 - a. Adrenal
 - b. Anterior pituitary
 - c. Posterior pituitary
 - d. Thyroid
6. Migrations change the number, distribution and composition of the population in
 - a. the area of departure
 - b. the area of arrival
 - c. both the area of departure and arrival
 - d. none of the above

Group 'B' [ZOOLOGY]

Give short answers to the following questions.

[4 × 4 = 16]

1. Give an account of the formation of coelom in the embryo of frog with labelled diagram.

[3+1]

OR

Write short notes on amniocentesis.

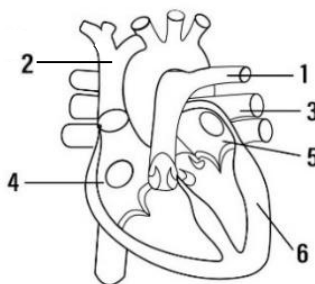
[4]

2. Overpopulation is a major issue in the development of the nation. Highlight the socio-economic problems caused by overpopulation and mention how to solve such problems.
3. Explain the organization of the mammary gland with the help of a diagram. What use does it off and when?
4. Identify the given figure. Give the names of labelled numbers in the given figure.

[2+2 = 4]

[3+1 = 4]

[1+3=4]



Group 'C' [ZOOLOGY]

Give long answers to the following questions.

[2 × 8 = 16]

5. Describe how urine is formed in the nephron through filtration, reabsorption and secretion. Draw well labelled diagram of nephron.

[3+2+2+1 = 8]

OR

Differentiate between sympathetic and parasympathetic nervous system. Describe the physiology of impulse conduction through nerve fiber.

[3+5 = 8]

6. Discuss about the candidiasis as you have studied.

[8]

ॐ The End ॐ